PERIDYNAMICS · REINFORCED CONCRETE

All results and model audit

Supplied Fortran reference, corrected rerun snapshots, and earlier independent Bowen reconstructions.

Report capture: 2026-10-10 10:09:26 (UTC+08:00)

What the results establish

The literal Fortran completed 100,000 steps, but its late response is physically invalid. A verified steel-yield force-component error explains approximately 1,300 m of steel motion. Its reported 1,059.81 kN reaction spike is not a valid structural prediction.

The error has been corrected in explicit variants. Seven native compiled regression tests pass. A corrected 200-step assembled check completed; two full corrected reruns were still in progress when this report was captured.

The supplied Fortran uses a different beam and loading case from Bowen Chapter 3. Neither it nor the earlier independent reconstructions establishes exact reproduction of 359.8 kN, 22.70 mm and the punching/shear crack pattern.

Result family	Status
Literal supplied Fortran	Complete numerical run; physically invalid late response
Corrected Fortran	Native tests pass; short run complete; full-run results preliminary
Earlier Bowen reconstructions	Documented finite-window results; author-model equivalence not demonstrated

This report includes all principal recorded results and comparisons. The complete literal time history and final nodal data are embedded as a ZIP attachment; active rerun data are frozen snapshots, not live values.

Download copy: embedded raw-data ZIP omitted to reduce file size.
Attachment instructions in this report apply to the full PDF only.

1. Which model was executed

The benchmark identity matters: the following setups describe different experiments and cannot be compared as interchangeable predictions.

Quantity	Supplied Fortran	Bowen Chapter 3
Beam / grid	Constructed length ≈3.458 m; node-centre extents 3.439 × 0.190 × 0.323 m	Physical beam 0.600 × 0.070 × 0.100 m
Supports	Reference x = ±1.5485 m; span 3.097 m	Span 0.500 m
Concrete / steel nodes	36,036 / 362	33,600 concrete; reinforcement discretization partly reconstructed
Grid / horizon	19 mm / 3.015 × grid	5 mm / 15 mm
Time step	3 microseconds	0.3 microseconds
Concrete modulus / density	30.5 GPa / 2500 kg/m³	25 GPa / 2400 kg/m³
Steel modulus / density	208 GPa / 7850 kg/m³	200 GPa / 7750 kg/m³
Reinforcement	Two straight 26 mm rods	Two 6 mm bars, with pictured end hooks
Loading	Two 10 mm diameter cylinders; prescribed 10 mm motion in 0.3 s	38.4 kg free impactor; initial speed 10.8 m/s; radius 60 mm
Constitutive thresholds	s0 = 5.00456e-5; sc = 1.25114e-3; steel sy = 0.002	Chapter 2 laws and Chapter 3 inputs; interface initialization partly missing

The original Fortran is self-contained and reads no external initialization files. Its representative steel volume is dx × 530.66 mm²; its concrete volume is dx³. The width/domain convention beyond outer node centres is not fully declared. No material inputs were fitted to target numbers.

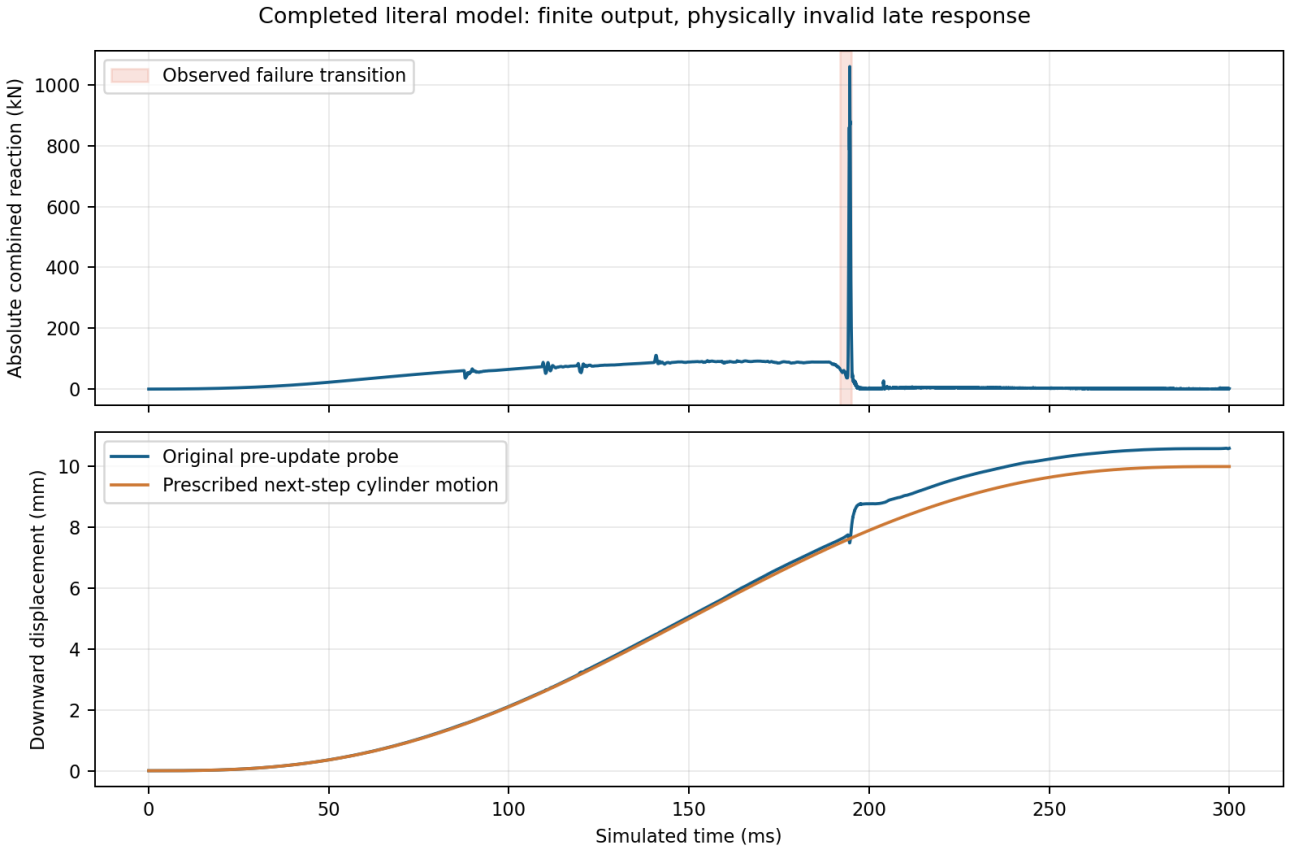
2. Completed literal run: numerical results

Recorded quantity	Value / time
Steps / simulated duration	100,000 / 0.300 s
Maximum absolute combined vertical contact reaction	1059.806437 kN; step 64,881; 194.643 ms
Maximum absolute original-probe displacement	10.602894 mm; step 99,720; 299.160 ms
Final prescribed cylinder displacement	10.000 mm downward
Final maximum steel displacement	1300.168100 m; node 181
Final maximum concrete displacement	1.494354 m; node 5171
Steel / concrete nodes above 100 mm motion	362 / 5,028
Final mean concrete damage indicator	0.806216
Final steel / concrete kinetic energy	1.5261075×10^{9} J / 0.557503 J
Finite output / solver exit	All history/state values finite; exit code 0
Physical assessment	Rejected as a structural prediction: catastrophic artificial steel forcing

The force is the sum of both cylindrical projection reactions. It is not Bowen's falling-hammer impact force. The original probe is at (0.0095, 0, −0.0095) m and is recorded before the current update; contact reaction is recorded after projection. These are not synchronous force–deflection observations.

The final displacement vector of steel node 181 is (−749.145, −751.627, −751.182) m. Norms in this audit are recomputed from actual vector components; the giant value is not a stale scalar-output artifact.

3. Literal histories and damage indicators



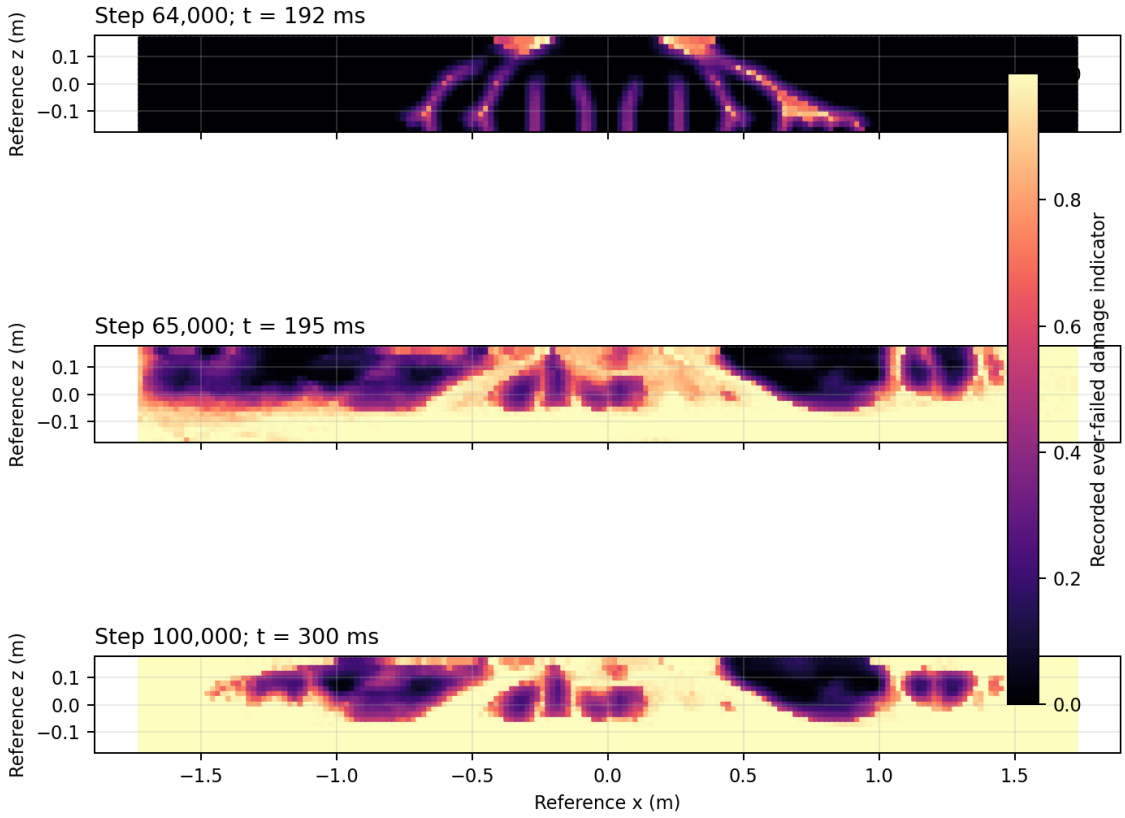
The red band marks the observed transition between recorded steps 64,000 and 65,000. Completion and finite arithmetic do not establish a valid response. The probe remains near 10 mm while steel and many concrete nodes depart far outside the beam.

Damage interpretation

The source stores ever-failed flags, but previous failure does not gate subsequent force evaluation. Its damage indicator therefore cannot establish permanent crack separation. Late damage images describe a corrupted computation; they are not evidence of Bowen's punching/shear crack pattern.

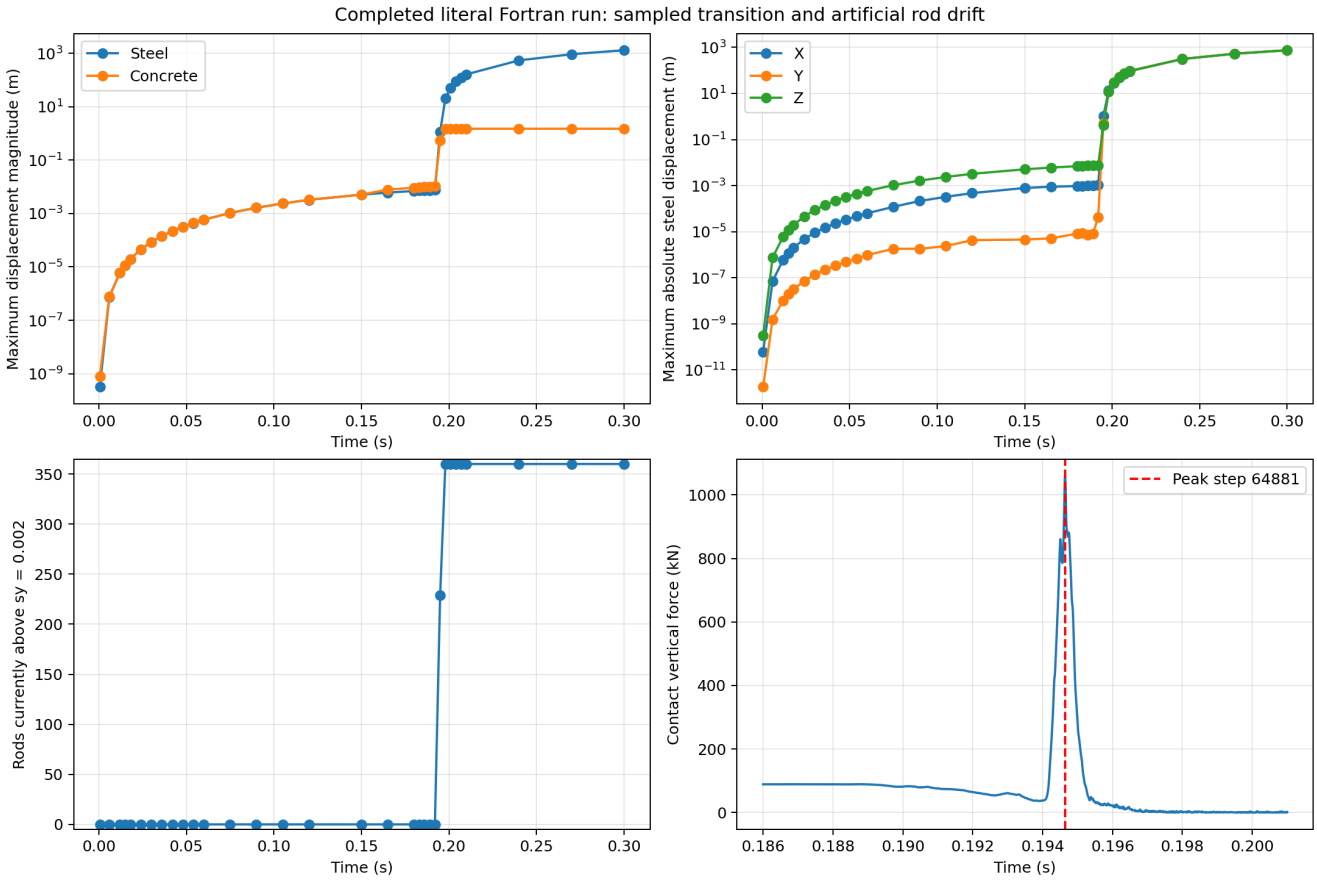
4. Reference-slice damage maps

Literal source damage in the y = 0 reference slice; late pattern is invalid



Colour is the original recorded concrete damage indicator on the y = 0 reference slice. Positions are reference coordinates, not the greatly displaced current geometry. The 195 ms and 300 ms states are affected by the identified runaway. The 192 ms state precedes the sampled transition but is not independently validated.

5. Failure trace and source of runaway



Step	Steel max motion	Concrete max motion	Rods above sy in snapshot
64,000	7.637 mm	10.418 mm	0
65,000	1.152 m	0.536 m	229
66,000	20.533 m	1.453 m	360
100,000	1300.168 m	1.494 m	360

The exact first yield event was not recorded in the original run. Snapshots bracket an observed transition, rather than proving the first branch activation. The 1,059.81 kN spike at step 64,881 lies within that transition.

6. Verified force-component error

Original source lines 746–747 copy the right endpoint's axial force accumulator into its y and z components:

```
pforce22(ccnode,2)=pforce22(ccnode,1)+0.0d0
pforce22(ccnode,3)=pforce22(ccnode,1)+0.0d0
```

The correction preserves each matching transverse accumulator. This matters for shared nodes: simply zeroing it would discard contributions from other elements.

```
pforce22(ccnode,2)=pforce22(ccnode,2)+0.0d0
pforce22(ccnode,3)=pforce22(ccnode,3)+0.0d0
```

One isolated yielded rod	Left force (N)	Right force (N)
Original	(220754.56, 0, 0)	(−220754.56, −220754.56, −220754.56)
Corrected	(220754.56, 0, 0)	(−220754.56, 0, 0)

With all 360 rods yielded, replaying the original accumulator order creates −79.4716416 MN net force in each transverse direction. Internal rod forces should sum to zero.

Quantitative fingerprint

Yield cap F = 416 × 530.66 = 220754.56 N. Declared steel volume Vs = 1.008254e−5 m³. Damping cn = 3e6. The erroneously forced terminal speed is:

```
|v_terminal| = (F / Vs) / cn
             = 7298.245614035088 m/s
```

Exact replay predicts every component of all 362 final steel velocities within 3.96e−10 m/s. In particular, 360 steel nodes reach the erroneous y/z terminal magnitude. This demonstrates sustained artificial forcing under damping, rather than merely an unexplained floating-point explosion. Reducing the timestep cannot remove that force.

7. Whole-model equation and code audit

Topic / source lines	Finding	Action / limitation
Steel forces; 710–750	Axial accumulator copied into transverse components after yield	Corrected component indices; axial law retained
PDROD; 779–818	Equal opposite densities do not give equal opposite physical forces for different volumes	Concrete return density × Vs/Vc; ratio 1.469972299
Acceleration; 896–926	pforce is N/m³; active update divides by density	Retain units; never divide the same density by nodal mass
Supports; 931–954	Clamped displacement but retained constrained velocity	Reset corresponding velocity at original support coordinates
Contact reaction	Concrete mass used even for contacted steel nodes	Use contacted node density and declared control volume
Cylinder tangent	y displacement/velocity reset despite x–z contact normal	Preserve y under declared frictionless-cylinder interpretation
Surface angle; 629–654	atan(dy/dx) can divide by zero	Explicit atan2 portability variant
Damage; 617–680	Prior failure flags do not gate force	Retained source history; permanent fracture not established
Rod consistency	Nonlinear length selects branch; elastic/plateau forces remain fixed-x	Finite-rotation objectivity issue remains; no silent constitutive replacement
Load output; 686–697	Top-layer reaction range is empty	Use cylindrical contact forceld rather than zero top-layer load columns
Observation; 699–703	Pre-update interior probe versus post-contact force	Preserve provenance; not identified with Bowen's sensor
Scalar motion; 956–958	Norm calculated before projection	Recompute from final vector components
Integration / energy	Damping-only and isolated rod checks do not prove full-system stability	Full convergence and energy ledger remain unverified
Benchmark identity	Long beam with prescribed cylinders differs from Chapter 3	Code corrections do not recover missing Chapter 3 definitions

No proven undefined initialization value explains the catastrophic motion. Compiler-specific behavior remains a provenance limitation. Initial rod-chain dt limit is approximately 3.689 microseconds; this is not a bound for the complete concrete/interface/contact system.

8. Corrections and numerical verification

All original moduli, densities, strengths, failure thresholds, damping, geometry, grid, timestep, load amplitude and load duration are retained. The corrections are selected explicitly and recorded in run manifests.

Native compiled regression	Result
Yielded rod physical action–reaction	PASS: artificial transverse resultant removed
Shared-node transverse accumulator	PASS: existing 17 N / 34 N contributions preserved
Pre-yield and compression branches	PASS: original and corrected values identical
Interface force and virtual work	PASS: mixed-volume physical balance restored
Stationary support velocities	PASS: corresponding constrained components reset
Contact reaction mass	PASS: concrete and steel momentum changes use their own masses
Frictionless cylinder tangent	PASS: nonzero y motion preserved; x–z projection unchanged

Tests compile actual extracted Fortran blocks. For a synthetic interface force-density vector, original physical net force is (3223.54, −6447.08, 9670.62) N; corrected residual is (0, 0, 3.64e−12) N. This tests implementation consistency for the declared volumes, not recovery of an unknown author calibration.

Completed 200-step assembled checks

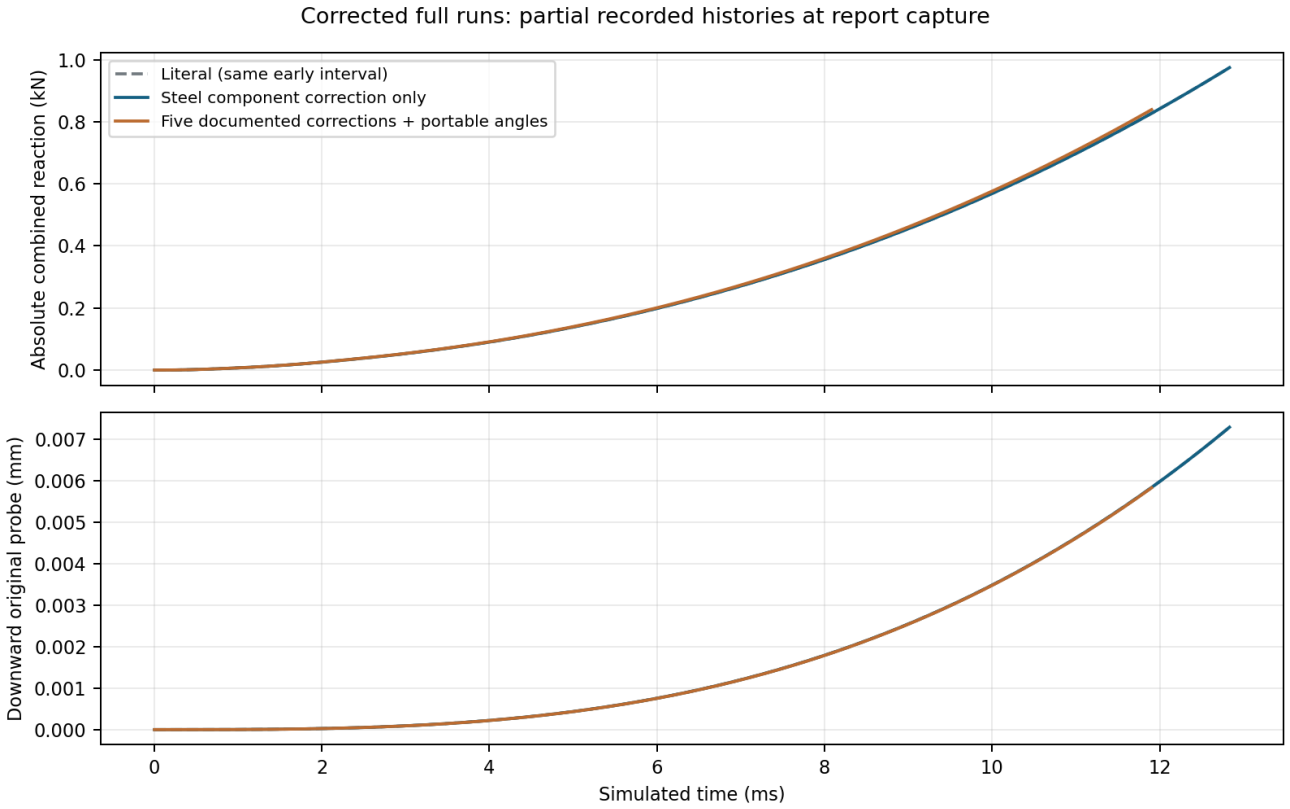
Variant	Duration	Observed max vertical reaction	Observed original probe
Literal source	0.600 ms	2.526034 N	3.183800e−7 mm
Five corrections + portable angles	0.600 ms	2.503090 N	3.083089e-07 mm

The corrected early run has zero recorded yielded rods and zero damage. Maximum logged interface resultant component is about 6.09e−16 N. These checks establish early assembled execution and force transfer; they do not establish the full late response. Earlier short checks did not activate the defective yield branch.

9. Full corrected reruns: report-time snapshot

Captured at 2026-10-10 10:09:26 (UTC+08:00). These are maxima observed within the completed portion only; final peaks are not yet available for an in-progress run.

Variant	Status / completed steps	Simulated time	Observed max force	Observed max probe
Steel component correction only	In progress 4,278 / 100,000	12.834 ms	0.974687 kN	0.007287 mm
Five documented corrections + portable angles	In progress 3,968 / 100,000	11.904 ms	0.839194 kN	0.005826 mm



The steel-only run isolates the two index corrections. The second run also balances interface physical force, resets constrained support velocities, uses appropriate contact masses, preserves cylinder tangential motion, and uses portable angles. Different stopping times are shown explicitly. Both preserve the complete 0.3 s loading schedule.

New diagnostics log steel/concrete maximum motion and speed, yielded rod count, net internal/interface forces, and the first yield event. The corrected model still retains source history and fixed-axis rod assumptions; successful completion alone will not validate it.

10. Earlier independent Bowen reconstructions

These fresh Python models are separate from the supplied Fortran and its corrected variants. They use a 0.60 m beam and Chapter 3 impact inputs, with explicitly unverified interface and setup choices. They are included for completeness, not combined with the Fortran benchmark.

Case	Window	Observed force	Displacement value
Bowen published simulation	Published full maxima	359.8 kN	22.70 mm
Published experiment	Published full maxima	301.0 kN	23.38 mm
Hooked, irreversible reconstruction	1.5 ms	220.662764 kN	16.547464 mm
Hooked, printed-history interpretation	1.5 ms	252.045237 kN	16.691092 mm
Irreversible, half timestep	0.6 ms	239.385821 kN	3.364648 mm

Computed displacement is the maximum observed bottom-layer midspan mean, not a source-matched experimental sensor. It was still increasing at 1.5 ms. Published maxima therefore cannot be treated as measurements over the same observation window.

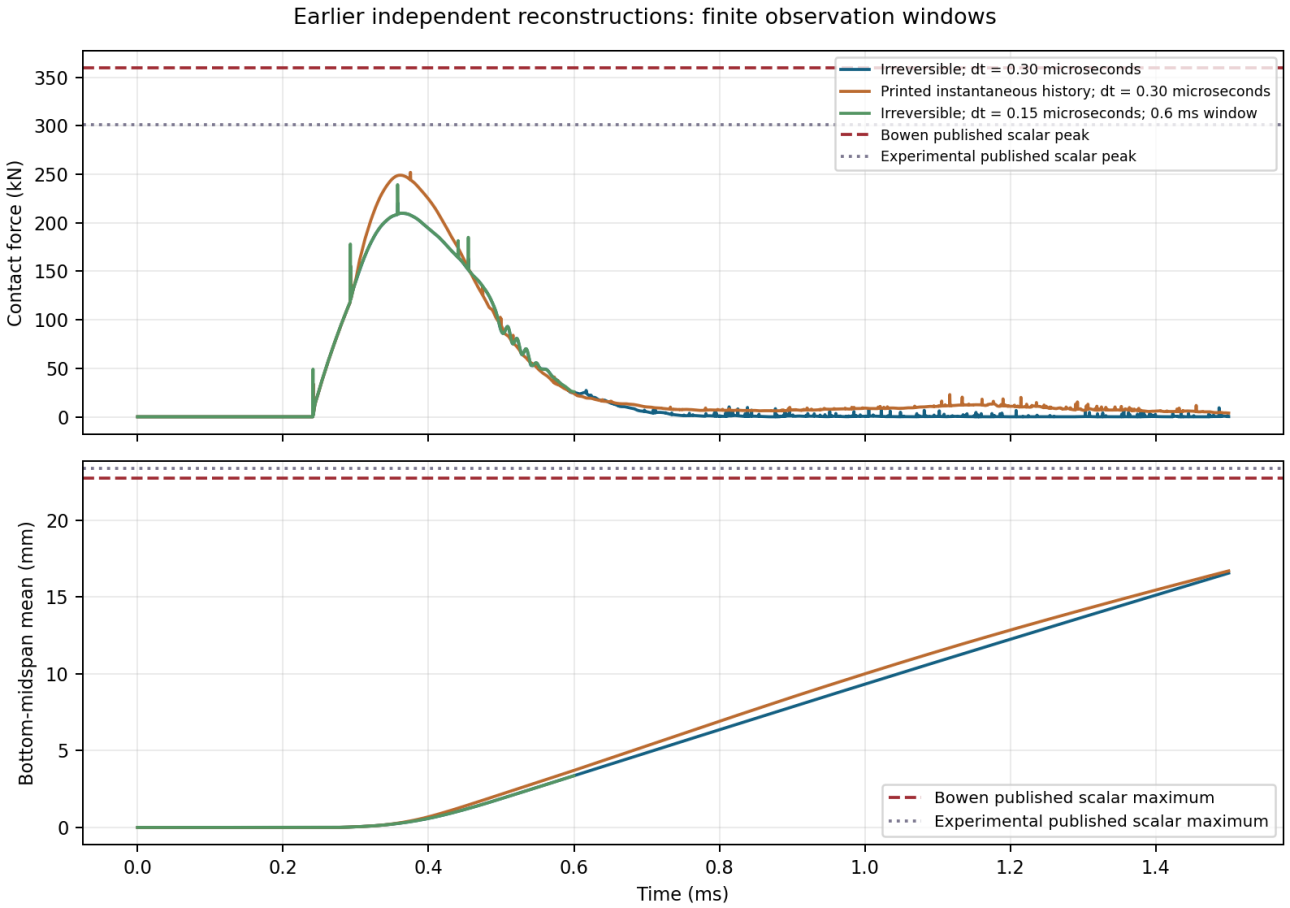
Diagnostic	Irreversible, 1.5 ms	Printed history, 1.5 ms
Mean damage indicator at window end	0.303813	0.318112
Max sampled closed-system material energy drift	2.025531 J	2.585204 J
Max momentum residual	6.5002e−10 kg·m/s	8.0105e−10 kg·m/s

These diagnostics use the earlier independent solver's definitions and are not a Fortran energy audit. Bowen's reported damage index 0.1701 has an incompletely identified reporting time; the above values do not form a matched-time comparison.

Timestep comparison at 0.6 ms

Halving the irreversible timestep changes the raw peak from 220.662764 to 239.385821 kN (+8.4849%). Contact impulse changes from 44.59855 to 44.66844 N·s (+0.1567%); end-window deflection changes from 3.368544 to 3.364648 mm (−0.1157%). An equivalent printed-history half-timestep run is absent. Raw peak convergence is unresolved.

11. Earlier reconstruction histories

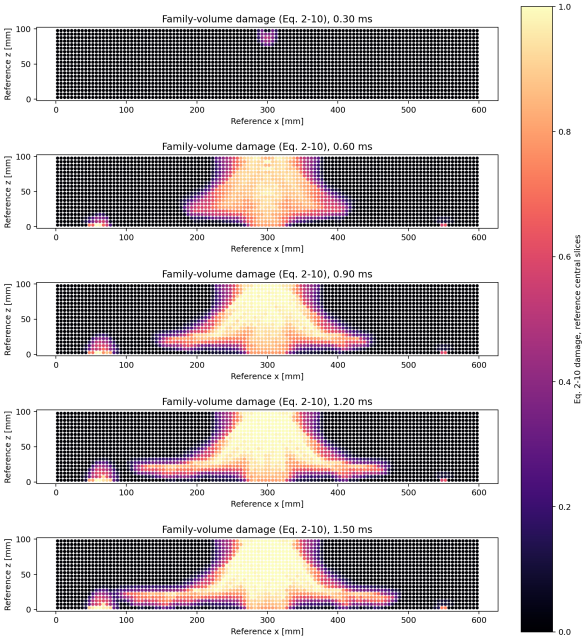


All three independent reconstructions are shown. Published values are scalar reference levels, not recovered original force–time or deflection–time curves. The half-timestep run stops at 0.6 ms; the two 1.5 ms deflections are still increasing. These observations do not establish a matched full-event maximum.

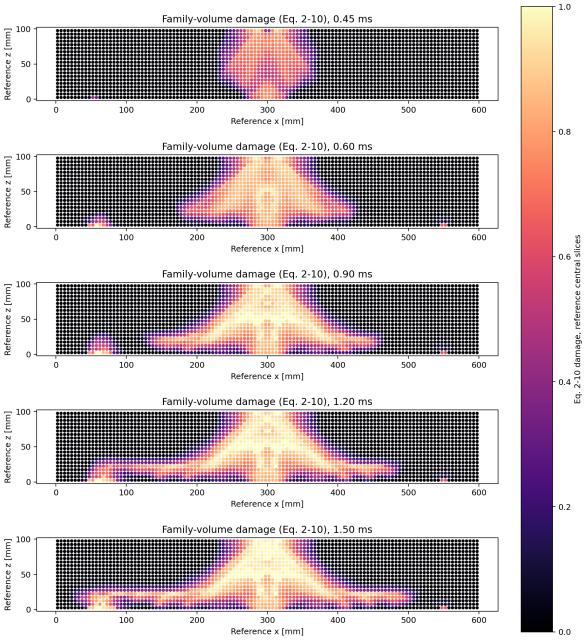
The raw-peak timestep difference is +8.4849% for the irreversible branch. The printed-force history can transmit force again after prior failure; its response and damage indicator require that history interpretation to remain explicit.

12. Earlier reconstruction damage figures

Irreversible history reconstruction



Printed instantaneous-force history reconstruction



Damage indicators in reference coordinates. Neither image establishes Bowen's punching/shear crack geometry. The printed-history image records ever-failed links, which may carry force again. Interface calibration, hook dimensions, boundary/contact details and original sensor definitions remain partly unverified.

13. Scope, outstanding checks and reproducibility

What is now demonstrated

A completed literal source calculation; a quantitatively identified steel force-component failure; explicit source corrections; seven passing compiled native regressions; finite early corrected execution; and documented partial histories for ongoing controlled full reruns.

What remains unresolved

Full corrected-run results, whole-event force/displacement convergence, a complete physical energy ledger, finite-deformation rod consistency, permanent fracture-history behavior, experimental validation and exact Chapter 3 equivalence. The literal reaction spike is rejected. Earlier reconstruction windows do not establish the published full-event maxima.

Embedded data

Open your PDF reader's attachments panel and extract **numeric_results_and_source_records.zip**. It contains the complete literal 100,000-step CSV, its 36,398-node final state, frozen partial corrected histories and diagnostics, audit data, test results, source/manifests, and earlier reconstruction records. The main PDF remains readable without extracting these files.

Recorded software and source identity

Compiler: GNU Fortran 14.2.0; literal build uses O2, bounds checks, backtraces, and invalid/overflow traps. Portable corrected run additionally traps division by zero. No fast-math compilation or material fitting is used.

Original source SHA-256:

`99d6a1b962a883d4ac181a23066ae140889cd53a957ebdb9050c155a376aaeca`

Native tests also record a separately named hash of the line-ending-normalized source. That hash differs from the immutable original byte hash because the supplied file uses CRLF.

Reproduce this report

```
cd /workspace/Khan/fortran_reference
../.venv/bin/python generate_results_pdf.py
../.venv/bin/python -m pytest audit/test_native_operators.py -q
```

Report generation freezes completed CSV rows at capture; running solvers are neither interrupted nor changed. Regenerating after reruns finish produces an updated report. Published Bowen/experimental targets come from the supplied thesis and earlier source audits; their raw original curves were not supplied.

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